

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Northern Bobwhite"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

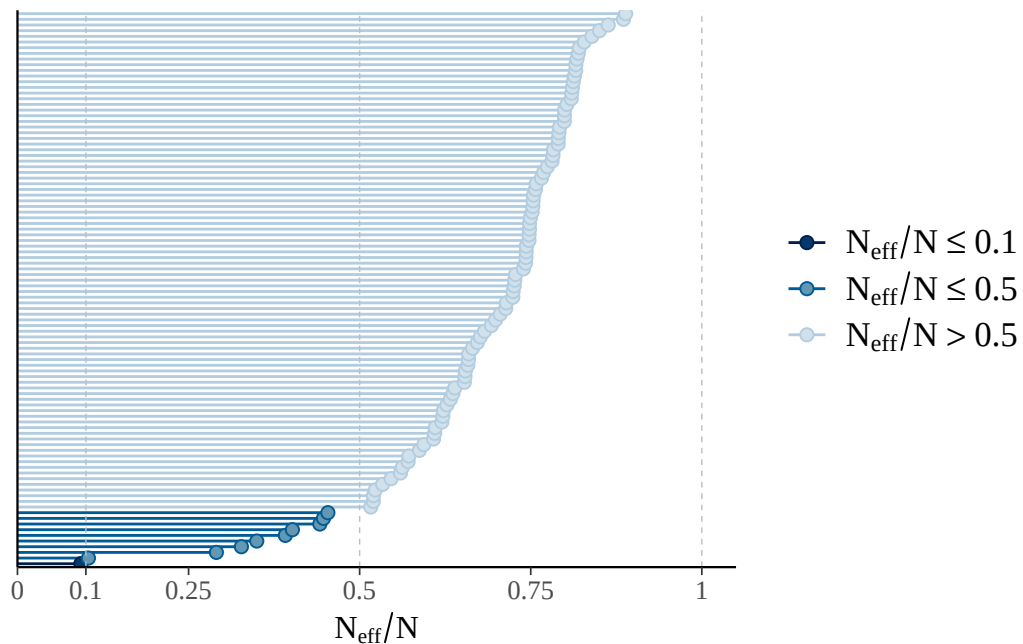
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.09272	0.65882	1.15710	1.10684	1.42872	2.40303

```
[1] "n_eff ratio tail ESS:"
```

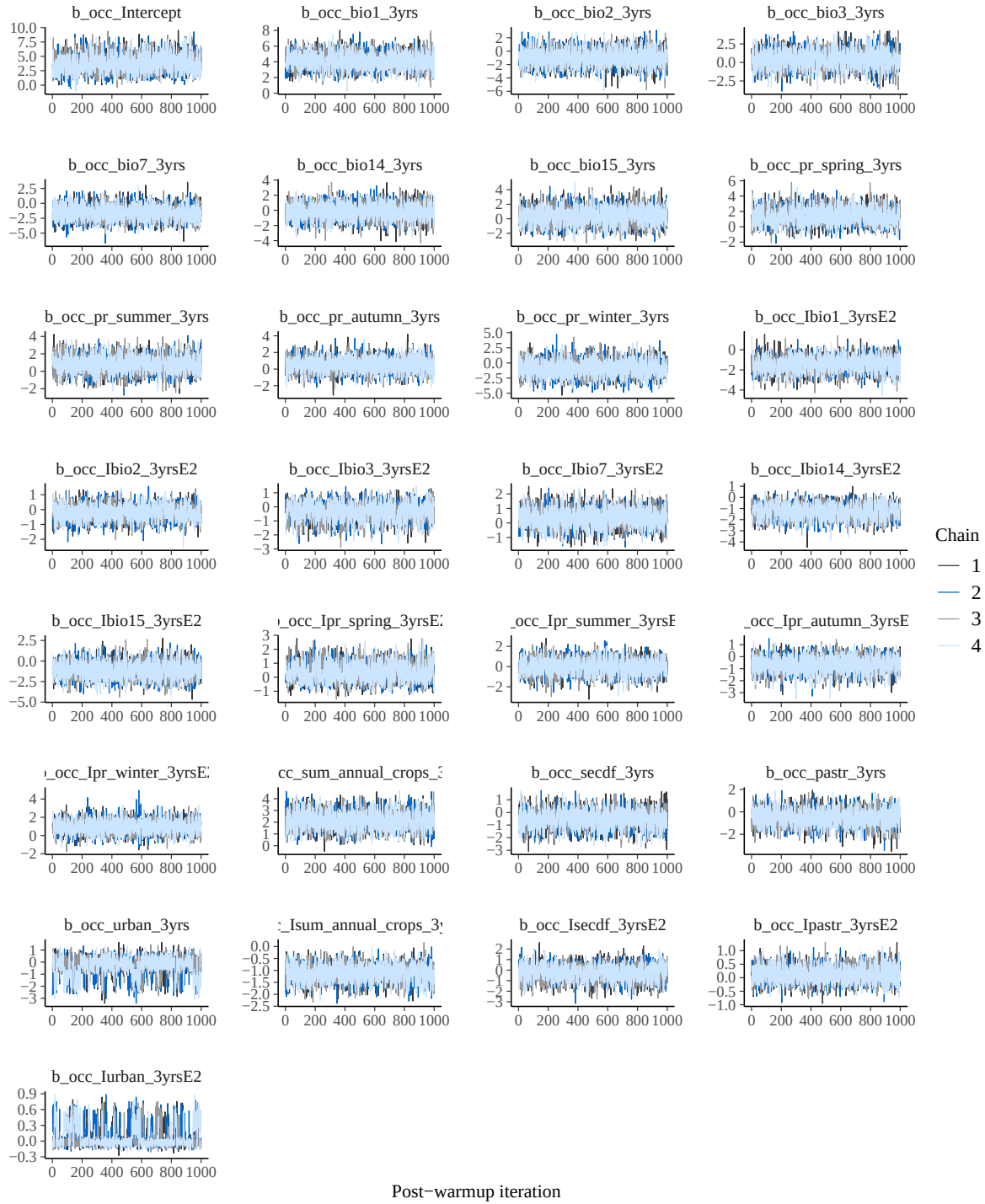
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2880	0.6543	0.7409	0.7152	0.7901	0.8887

```
[1] "effective sample size in bulk or tail too small"
```

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.

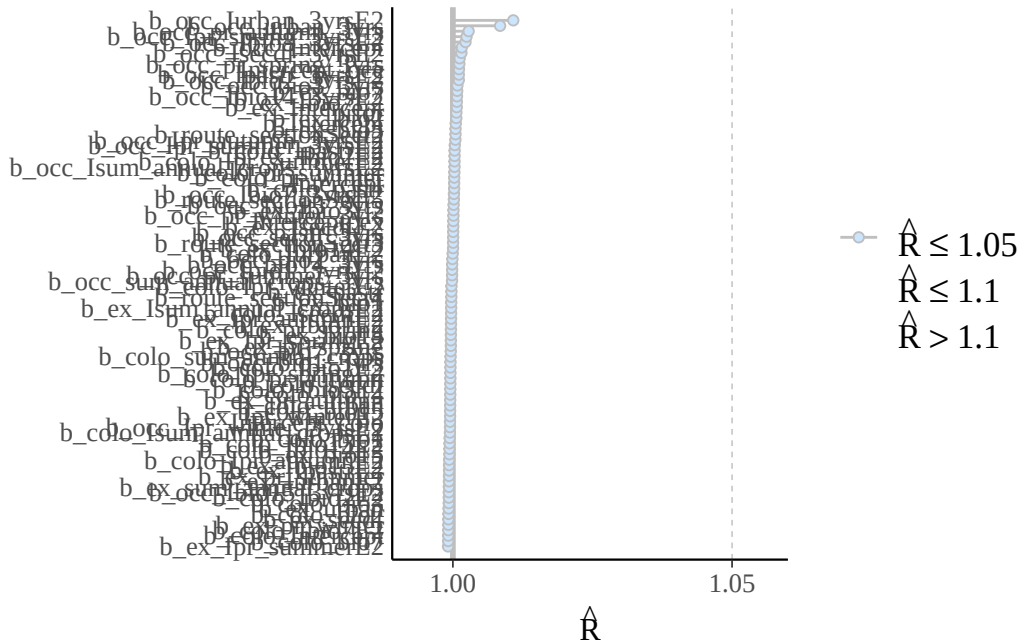


Post-warmup iteration



Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9995	0.9998	1.0002	1.0003	1.0108



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"          "b_occ_bio1_3yrs"
[3] "b_occ_Ibio14_3yrsE2"     "b_occ_sum_annual_crops_3yrs"
[5] "b_occ_Isum_annual_crops_3yrsE2"
```

```
[1] "b_colo_bio1"          "b_colo_Ibio1E2"
[3] "b_colo_sum_annual_crops" "b_colo_urban"
[5] "b_colo_Isum_annual_cropsE2"
```

```
[1] "b_ex_Intercept"      "b_ex_bio1"
[3] "b_ex_bio2"           "b_ex_bio3"
[5] "b_ex_bio7"           "b_ex_pr_summer"
[7] "b_ex_pr_winter"      "b_ex_Ibio1E2"
```

```
[9] "b_ex_Ibio2E2"          "b_ex_Ibio3E2"
[11] "b_ex_Ibio7E2"          "b_ex_sum_annual_crops"
[13] "b_ex_secdf"            "b_ex_urban"
[15] "b_ex_Isum_annual_cropsE2" "b_ex_IsecdfE2"
[17] "b_ex_IpastrE2"
```

